

BOUDROUSS RÉDA | CV

Software Engineer – OCaml, Programming Languages & Static Analysis – Paris

github.com/rboudrouss
linkedin.com/in/rboudrouss
contact@rboud.com
rboud.com
+33 (0)7 55 78 85 16

Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology, 2026) at Sorbonne University, specialized in the design and implementation of **programming languages**, **static typing** and **abstract interpretation**. I am most at home in **OCaml**, and ported the MOPSA static analyzer (OCaml/C/C++) to **WebAssembly**. At Hexaglobe I built test strategies, CI/CD and AI-agent tooling spanning four stacks. I like building robust, well-typed software across the stack, from language internals to UI.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc

2026

- Ported **MOPSA**, an **abstract-interpretation** static analyzer for C and Python written in **OCaml/C/C++**, to **WebAssembly**, statically linking the OCaml runtime, the Clang frontend and the Apron numerical domains with their GMP/MPFR arithmetic back-ends using Emscripten.
- Dug into the **OCaml runtime and C FFI**, statically registering ~1,400 C primitives and root-causing an unsigned-wraparound bug in the Tag_val macro. Recompiled Apron to use exact GMP rationals, addressing the soundness hazard of Wasm's fixed rounding mode.
- Shipped a fully client-side **playground** (interactive REPL over SharedArrayBuffer + Atomics) and a detailed technical **write-up**.

Reactant analyzer - github.com/rboudrouss/reactant-analyzer

2026

- Designed and implemented an **abstract-interpretation**-based static analyzer in **Rust** (~30k lines, 500+ tests) that catches **React** hook bugs beyond the reach of syntactic linters, such as infinite render loops, derived state and unstable dependencies.
- Evaluates components in abstract domains (intervals, constant sets, reference-stability tags) over a dedicated CFG-based IR and flags an infinite render loop when the state fixpoint **widens**. Ships as a CLI with JSON output for CI and a Rust plugin API.

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University

2024 – 2026

- Coursework: design & implementation of **programming languages** (interpreters, compilers), formal semantics, **type systems** and static analysis by abstract interpretation (course led by Antoine Miné), advanced **functional programming**, advanced algorithmics, concurrent & distributed programming.

BSc in Computer Science - Sorbonne University

2020 – 2024

- Algorithms, data structures, computer architecture, operating systems, networks and databases.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming)

2025 – 2026

- Designed complete **test strategies** (unit, integration, E2E Playwright) for zero-coverage legacy products, reaching ~**3,000 test cases** across 4 stacks (TypeScript, Python, PHP, JS).
- Industrialized test production with **AI coding agents** (Claude Code), combining sandboxed planner/implementer/reviewer orchestration, versioned guardrails, mandatory green tests and systematic human review to produce ~720 integration tests in 3 days.
- Built objective **quality oracles** (VMAF, black-frame/freeze detection, audio analysis) to validate a live video encoder, and **GitLab CI/CD** pipelines with shared CI components consumed by 4+ repositories.

Technical Lead, then President (2024–25) - Association Play Sorbonne Université

2023 – 2026

- Built a **local-first** mobile web app (secure authentication, backend optimized for low-bandwidth clients) and the association's website, both running on a virtualized infrastructure I operated (Docker servers, CI/CD).
- Led **100+ volunteers** as President for a cultural event welcoming 10,000+ visitors.